

AWS IAM Security Audit – Documentation

1. Project Title

AWS IAM Security Audit Using Users, Groups, Policies, Access Analyzer, and CloudTrail

2. Objective

The objective of this project is to implement and audit AWS Identity and Access Management (IAM) security controls. The project demonstrates secure user management, group-based permission assignment, custom policy creation, multi-factor authentication, password security configuration, access restriction testing, and activity monitoring using AWS CloudTrail.

The implementation focuses on applying the principle of least privilege by creating IAM users, assigning permissions through groups and policies, and verifying authorized and unauthorized access attempts.

3. AWS Services Used

- AWS IAM (Identity and Access Management)
- IAM Access Analyzer
- AWS CloudTrail
- Amazon S3

4. Project Details

IAM Users Created

The following IAM users were created for testing different access levels:

User	Purpose
security-admin	Security administration activities
developer-user	Developer-level resource access

audit-user	Audit and monitoring activities
------------	---------------------------------

IAM Groups Created

IAM groups were created to manage permissions efficiently.

Group	Purpose
Admins	Provides security management permissions
Developers	Provides development-related permissions
Auditors	Provides read-only auditing permissions

IAM Policies Configured

Custom IAM policies were created and attached to users/groups based on their responsibilities.

Implemented policies include:

- Developer access policy
- Security administration policy
- Audit access policy

Inline policies were also configured to provide specific permissions required for users.

Security Configurations Implemented

Password Policy

Configured AWS account password requirements:

- Strong password requirements
- Password expiration settings
- Password reuse prevention

Multi-Factor Authentication (MFA)

Enabled MFA authentication for IAM users to provide an additional layer of account security.

Access Testing

The following access tests were performed:

Authorized Access:

- Developer-user successfully accessed allowed resources.
- S3 object upload was tested successfully.

Unauthorized Access:

- Restricted actions were tested.
- Access Denied responses were verified.

IAM Access Analyzer Configuration

Analyzer Name:

IAM-Security-Analyzer

Finding Type:

Resource Analysis - External Access

IAM Access Analyzer was configured to identify resources that may allow unintended external access.

CloudTrail Configuration

Trail Name:

iam-security-audit-trail-smrithi

Event Logging:

Management Events

Enabled:

- Read API activity
- Write API activity

Log Storage:

Amazon S3

CloudTrail was configured to record AWS account activities for auditing purposes.

5. Project Architecture



6. Steps Followed to Implement IAM Security Audit

Step 1: Create IAM Users

1. Opened AWS Management Console.
2. Navigated to:

IAM → Users

3. Created IAM users:
 - security-admin
 - developer-user
 - audit-user
4. Configured user access credentials.

Step 2: Create IAM Groups

1. Navigated to:

IAM → User Groups

2. Created groups:
 - Security-Admins
 - Developers
 - Auditors
3. Groups were used to simplify permission management.
4. Users were assigned to their respective groups.

Step 3: Create Custom IAM Policies

1. Opened:

IAM → Policies

2. Created custom policies according to user requirements.
3. Defined permissions using JSON policy documents.
4. Attached policies to appropriate IAM groups/users.

This implemented controlled access management.

Step 4: Configure Inline Policies

1. Selected required IAM users/groups.
2. Added inline policies.
3. Defined specific permissions required for individual users.

Inline policies were used for user-specific access control.

Step 5: Configure IAM Password Policy

1. Opened:

IAM → Account Settings

2. Configured password policy settings.

Implemented:

- Password complexity requirements
- Password expiration
- Password history restrictions

Step 6: Enable MFA Authentication

1. Opened IAM user security credentials.
2. Selected MFA device configuration.
3. Registered an MFA device.
4. Verified successful MFA authentication.

MFA was enabled to improve account security.

Step 7: Test IAM User Access

Developer User Testing

1. Logged in using developer-user credentials.
2. Access allowed AWS services.
3. Uploaded objects to an S3 bucket.

Result:

Access Successful

Unauthorized Access Testing

1. Attempted restricted operations.
2. Verified AWS denied the request.

Result:

Access Denied

This confirmed that least privilege permissions were working correctly.

Step 8: Configure IAM Access Analyzer

1. Opened:

IAM → Access Analyzer

2. Created a new analyzer.
3. Selected:

Resource Analysis - External Access

4. Enabled analysis of resource permissions.

Step 9: Configure AWS CloudTrail

1. Opened:

CloudTrail → Trails

2. Created a new trail.

Trail Name:

iam-security-audit-trail-smrithi

3. Enabled management events.
4. Configured S3 as the log storage destination.

CloudTrail was used to record IAM and AWS resource activities.

Step 10: Verify CloudTrail Events

CloudTrail Event History was checked to verify recorded actions.

Observed activities included:

- IAM operations
- S3 operations
- CloudTrail events
- User activity tracking

The logs confirmed successful auditing of AWS account activity.

7. Security Controls Implemented

Security Control	Implementation
User Management	IAM Users
Permission Management	IAM Groups and Policies

Authentication Security	MFA
Password Security	IAM Password Policy
Least Privilege	Custom and Inline Policies
Access Monitoring	AWS CloudTrail
External Access Detection	IAM Access Analyzer
Resource Testing	S3 Access Validation

8. Output

- Successfully created IAM users and groups.
- Configured custom IAM policies and inline policies.
- Implemented password security settings.
- Enabled MFA authentication.
- Verified authorized and denied access scenarios.
- Configured IAM Access Analyzer.
- Enabled CloudTrail auditing.
- Verified AWS activities through CloudTrail Event History.

9. Conclusion

This project demonstrates the implementation of AWS IAM security auditing practices using IAM users, groups, policies, IAM Access Analyzer, and CloudTrail.

The project successfully implemented secure identity management, controlled resource access, least privilege permissions, MFA authentication, and activity monitoring. The configured environment helps improve AWS security posture by controlling user permissions and tracking account activities.